

TrES-5b

R-1072

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_160508 · created 2026-07-30T13:59:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457516.9162 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.025
Transit depth (Rp/Rs)^2	2.59 +/- 0.81 [%]
Orbital Inclination (inc)	83.75 +/- 2.0
Airmass coefficient 1 (a1)	0.989 +/- 0.022
Airmass coefficient 2 (a2)	0.011 +/- 0.012
Scatter in the residuals of the lightcurve fit is	2.19 %
Best Comparison Star	#1 - [36, 98]
Optimal Method	PSF photometry
Transit Duration (day)	0.069 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.161 ± 0.025 vs 0.142 (deviation 0.54σ)
Duration fit vs published	0.069 d vs 0.0758 d (deviation 0.3σ)
Mid-time O–C	-0.1 min
Depth SNR	4.6
Residual scatter	2.19 %

Light curve

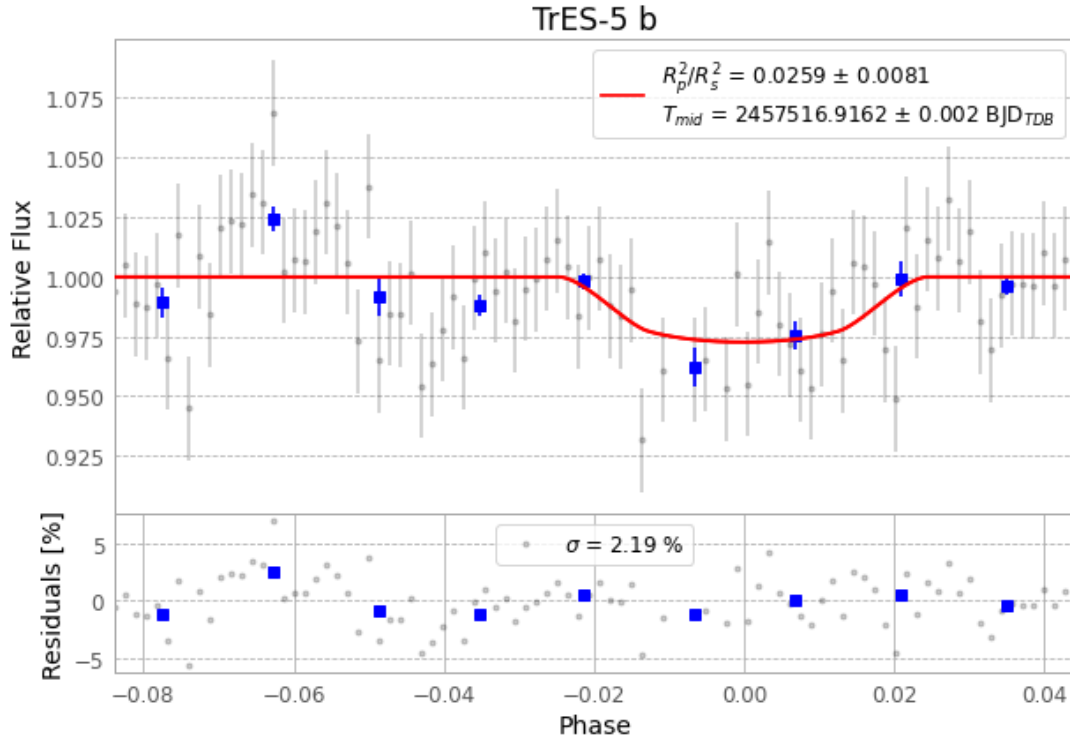


Figure 1 — FinalLightCurve_TrES-5 b_2016-05-08.png (SHA-256 34155a961f77243b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 245], 1 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 5d7fc80a3c17de31...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

92 FITS frames (60 MB), TRES-5160508070012.FITS → TRES-5160508113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-160508.log (4d6c3ef4d9a6...), FinalLightCurve_TrES-5 b_2016-05-08.png (34155a961f77...), FinalLightCurve_TrES-5 b_2016-05-08.csv (2640d3c897f4...)

Compute resources

Wall time 70.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 13:59Z.